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Driving assembly and transfer device

Abstract

A driving assembly and a transfer device to allow an adjustable length of wheelbase for varying sizes of loads comprises a driving device, a driving wheel connected to the driving device rotatable by the driving device, a cantilever assembly comprising a cantilever configured for mounting to the bearing platform, and a driven wheel rotatably connected to one end of the cantilever. The other end of the cantilever is detachably connected to a fixed part of the driving device. The transfer device comprises a bearing platform and two driving assemblies.

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Background/Summary

FIELD

(1) The subject herein generally relates to transfer device, and more particularly to a driving assembly and a transfer device.

BACKGROUND

(2) Most existing transfer platforms adopt an integrated design, and structures with a suitable bearing capacity are required for different loads, so the components of these transfer platforms have poor versatility.

SUMMARY OF THE INVENTION

(3) The present invention is to provide a driving assembly and a transfer device.

(4) In a first aspect, the disclosure provides a driving assembly for driving a bearing platform, the driving assembly comprises: a driving device; a driving wheel connected to the driving device, rotatable by the driving device; a cantilever assembly comprising a cantilever configured for mounting to the bearing platform; and, a driven wheel rotatably connected to one end of the cantilever, and the other end of the cantilever is detachably connected to a fixed part of the driving device.

(5) Optionally, the driving device further comprises a driving component and a mounting plate. The mounting plate comprises opposite first and second ends, the first end is fixedly connected with the driving component, and the second end protrudes from the driving component along a first direction.

(6) Optionally, the cantilever successively comprises a first portion, a second portion, and a third portion in the first direction. Wherein the first portion is connected with the second end, the third portion is connected with the driven wheel, the second portion is connected between the first portion and the third portion, the third portion and the first portion extend along the first direction respectively and are arranged at parallel intervals in the third direction, wherein the third direction is perpendicular to the first direction.

(7) Optionally, a first bump is provided at one end of the first portion along the second direction, a second bump is provided at another end of the first portion along the second direction, the first bump and the second bump are arranged on the same side of the first portion along the first direction, and the second direction is perpendicular to the first direction. The first bump is detachably connected with the second end along the second direction.

(8) Optionally, a plurality of positioning columns and a plurality of first screw holes are provided at top surface along the second direction of the first bump. The mounting plate is provided with a plurality of positioning holes matched with the positioning column and a plurality of through holes corresponding to the first screw hole. The first bump is detachably connected to the second portion by bolts passing through the first screw holes and the through holes.

(9) Optionally, the second bump is provided with a first connecting opening, a second connecting opening, and a third connecting opening along the second direction. The second portion is provided with a first inserting block, a second inserting block, and a third inserting block respectively matched to the first connecting opening, the second connecting opening, and the third connecting opening. The first inserting block is inserted into the first connecting opening, the second inserting block is inserted into the second connecting opening, and the third inserting block is inserted into the third connecting opening, to connect the second bump and the second portion together.

(10) Optionally, the driving assembly further comprises a bearing frame connected to the bearing platform along the second direction. The bearing frame comprises a fixing plate and a support shaft, the fixing plate is connected to the bearing platform, one end of the support shaft is connected with the fixing plate along the second direction, and middle part of the cantilever is rotatably supported on the support shaft.

(11) In a second aspect, the disclosure provides a transfer device, wherein the transfer device comprises the bearing platform and two driving assemblies described above, the two driving assemblies are arranged on opposite sides of the bearing platform along the first direction, wherein the two driving assemblies or the cantilever is connected with the bearing platform.

(12) Optionally, the driving device further comprises a driving component and a mounting plate. The mounting plate comprises opposite first and second ends, the first end is fixedly connected with the driving component, and the second end protrudes from the driving component along a first

direction.

(13) Optionally, the cantilever comprises a first portion, a second portion, and a third portion in the first direction. Wherein the first portion is connected with the second end, the third portion is connected with the driven wheel, the second portion is connected between the first portion and the third portion, the third portion and the first portion extend along the first direction respectively and are arranged at parallel intervals in the third direction, wherein the third direction is perpendicular to the first direction.

(14) Optionally, a first bump is provided at one end of the first portion along the second direction, a second bump is provided at another end of the first portion along the second direction, the first bump and the second bump are arranged on the same side of the first portion along the first direction, and the second direction is perpendicular to the first direction. The first bump is detachably connected with the second end along the second direction.

(15) Optionally, a plurality of positioning columns and a plurality of first screw holes are provided at top surface along the second direction of the first bump. The mounting plate is provided with a plurality of positioning holes matched with the positioning column and a plurality of through holes corresponding to the first screw hole. The first bump is detachably connected to the second portion by bolts passing through the first screw holes and the through holes.

(16) Optionally, the second bump is provided with a first connecting opening, a second connecting opening, and a third connecting opening along the second direction. The second portion is provided with a first inserting block, a second inserting block, and a third inserting block respectively matched to the first connecting opening, the second connecting opening, and the third connecting opening. The first inserting block is inserted into the first connecting opening, the second inserting block is inserted into the second connecting opening, and the third inserting block is inserted into the third connecting opening, to connect the second bump and the second portion together.

(17) Optionally, the driving assembly further comprises a bearing frame connected to the bearing platform along the second direction. The bearing frame comprises a fixing plate and a support shaft, the fixing plate is connected to the bearing platform, one end of the support shaft is connected with the fixing plate along the second direction, and middle part of the cantilever is rotatably supported on the support shaft.

(18) Optionally, the transfer device further comprises a connecting bridge and two secondary driven wheels, the two secondary driven wheels are rotatably connected to both ends of the connecting bridge respectively. A middle of the connecting bridge is rotatably connected to the bearing platform, and the two secondary driven wheels are located at an end of the bearing platform opposite to the driven wheel along the first direction.

(19) Optionally, the driven wheel and the secondary driven wheel are omni-directional wheels.

(20) Compared with the prior art, the driving assembly in this embodiment is detachably connected with the driving device through the cantilever. By using cantilevers of different lengths, the driving device with different wheelbases can be obtained, so as to realize the sharing of parts, reducing costs, and facilitating maintenance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to more clearly explain the technical scheme of the disclosure, the accompanying drawings in the embodiment will be briefly introduced below. It should be understood that the following accompanying drawings only show some and not all embodiments of the invention, so they should not be regarded as limiting the scope. For those skilled in the art, without paying creative labor, relevant drawings can also be developed from these drawings.

(2) FIG. 1 is a schematic diagram of a driving assembly according to an embodiment of the present

application.

(3) FIG. 2 is a schematic diagram of the mounting plate in the driving assembly according to an embodiment of the present application.

(4) FIG. 3 is a schematic diagram of a cantilever assembly in the driving assembly according to an embodiment of the present application.

(5) FIG. 4 is a schematic diagram of a cantilever in the driving assembly according to an embodiment of the present application.

(6) FIG. 5 is an exploded view of the cantilever in the driving assembly according to an embodiment of the present application.

(7) FIG. 6 is a schematic diagram of a driving wheel in the driving assembly according to an embodiment of the present application.

(8) FIG. 7 is a schematic diagram of the transfer device according to an embodiment of the present application.

(9) FIG. 8 is another view of the transfer device of FIG. 7.

DESCRIPTION OF MAIN COMPONENT SYMBOLS

(10) driving assembly **100** driving wheel **10a** driven wheel **10b** secondary driven wheel **10c** shaft hole **101** cantilever assembly **12** cantilever **120** first portion **120a** second portion **120b** third portion **120c** first connecting opening **1203** second connecting opening **1204** third connecting opening **1205** first inserting block **1206** second inserting block **1207** third inserting block **1208** first bump **121a** second bump **121b** receiving cavity **122** first cavity **122a** second cavity **122b** first annular boss **122c** positioning column **123** first screw hole **124** bearing **125** support frame **126** fixing plate **126a** support shaft **126b** screw **128** nut **129a** washer **129b** driving device **13** driving component **13a** mounting plate **13b** first end **130a** second end **130b** mounting hole **131** connecting hole **132** positioning hole **133** through hole **134** second annular boss **135** curved through slot **136** flange **14** body **14a** annular protrusion **14b** second screw hole **141** third screw hole **142** transfer device **200** bearing platform **20** connecting bridge **21** bearing assembly **22** sensing module **23**

DETAILED DESCRIPTION

(11) The technical scheme in the embodiment of the application will be described in combination with the attached drawings in the embodiment of the application. Obviously, the described embodiments are only part of the embodiments of the application, not all embodiments.

(12) It should be noted that when an element is said to be “fixed to” another element, it can be directly on another element or there can be an intervening element. When a component is “connected” to another component, it can be directly connected to another component or there may be intermediate components at the same time. When a component is “set on” another component, it can be set directly on another component or there may be intervening components at the same time. The terms “vertical”, “horizontal”, “left”, “right” and similar expressions used in this paper are for illustrative purposes only.

(13) Unless otherwise defined, all technical and scientific terms used herein have the same meanings as those commonly understood by those skilled in the technical field of the application. The terms used in the specification of the application herein are only for the purpose of describing specific embodiments and are not intended to limit the application. The term “and/or” as used herein includes any and all combinations of one or more related listed items.

(14) Some embodiments of the present application are described in detail. Without conflict, the following embodiments and features in the embodiments can be combined with each other.

(15) The specific embodiments of the present application will be further described in detail below with reference to the accompanying drawings.

(16) Referring to FIGS. 1, 3 and 7, X axis in the drawings is named as the first direction, Y axis in the drawings is named as the second direction, and Z axis in the drawings is named as the third direction.

(17) The embodiment of the application provides driving assembly **100** for driving a bearing

platform **20**. The driving assembly **100** includes driving device **13**, driving wheel **10a**, cantilever assembly **12**, and driven wheel **10b**. The driving wheel **10a** is connected to the driving device **13** and rotates when driven by the driving device **13**. The cantilever assembly **12** includes a cantilever **120**. The driven wheel **10b** is rotatably connected to one end of the cantilever **120**, and the other end of the cantilever **120** is detachably connected to the fixed part **13c** of the driving device **13**. The driving device **13** may be an electric motor. The electric motor includes a housing for mounting the electric motor to a base and an output shaft that can rotate when energized.

(18) When in use, the driving assembly **100** in this embodiment is detachably connected with the driving device **13** through the cantilever **120**. By using cantilevers of different lengths, the driving device **13** with different wheelbases can be obtained, so as to realize the sharing of parts, reducing costs, and facilitating maintenance.

(19) Referring to FIGS. **1** and **2**, the driving device **13** further comprises a driving component **13a** and a mounting plate **13b**. The mounting plate **13b** includes opposite first end **130a** and second end **130b**, the first end **130a** is fixedly connected with the driving component **13a**, and the second end **130b** protrudes from the driving component **13a** in the first direction. It should be noted that the fixed part **13c** of the driving device **13** can be the mounting plate **13b** or a housing of the driving component **13a**.

(20) Referring to FIGS. **2** and **6**, the first end **130a** is provided with a mounting hole **131**, and a plurality of connecting holes **132** are arranged around the mounting hole **131**. The driving wheel **10a** is provided with a flange **14** which is fixedly connected with the driving wheel **10a**. The flange **14** includes a body **14a** and an annular protrusion **14b**. The body **14a** is provided with a second screw hole **141** corresponding to the connecting hole **132**. The mounting plate **13b** is sleeved on the outside of the annular protrusion **14b** through the mounting hole **131**, and the mounting plate **13b** is fixedly connected with the flange **14** through a bolt (not shown in the drawings) through the connecting hole **132** and the second screw hole **141**.

(21) Further, the first end **130a** is also provided with a second annular boss **135**, the second annular boss **135** is arranged on the inner surface of the mounting hole **131**, and the second annular boss **135** is butted with the annular boss **14b** in the second direction. The second direction is perpendicular to the first direction.

(22) Further, the inner surface of the second annular boss **135** is provided with a plurality of curved through slots **136**, which penetrate the second annular boss **135**. The top surface of the annular protrusion **14b** is provided with third screw holes **142** at intervals, and the third screw holes **142** correspond one-to-one with the curved through slots **136**. The driving component **13a** is provided with an extended output shaft (not shown in the drawings), and the driving wheel **10a** is provided with a shaft hole **101** adapted to the output shaft. The output shaft is inserted into the shaft hole **101** to connect with the driving wheel **10a**, and the driving component **13a** is fixedly connected with the flange **14** by the bolt through the curved through slots **136** and the third screw holes **142**. The rotation of the mounting plate **13b** is prevented by matching the bolt with the curved through slots **136**.

(23) Referring to FIG. **1**, the cantilever **120** includes a first portion **120a**, a second portion **120b**, and a third portion **120c** in the first direction. Wherein, the first portion **120a** is connected with the second end **130b**, the third portion **120c** is connected with the driven wheel **10b**, the second portion **120b** is connected between the first portion **120a** and the third portion **120c**, the third portion **120c** and the first portion **120a** extend along the first direction respectively and are arranged to be parallel at intervals in the third direction, wherein the third direction is perpendicular to the first direction.

(24) Referring to FIG. **4**, the first portion **120a** is provided with a first bump **121a** and a second bump **121b** at both ends in the second direction. The first bump **121a** and the second bump **121b** are arranged on the same side of the first portion **120a** in the first direction so that the first bump **121a**, the second bump **121b** and the first portion **120a** are connected to form a U-shaped structure.

The first bump **121a** is detachably connected with the second end **130b** in the second direction.

(25) Further, referring to FIGS. **4** and **5**, the top surface of the first bump **121a** along the second direction is provided with a plurality of positioning columns **123** and a plurality of first screw holes **124**. The mounting plate **13b** is provided with a plurality of positioning holes **133** matched with the positioning column **123** and a plurality of through holes **134** corresponding to the first screw holes **124**. The first bump **121a** is detachably connected with the second end **130b** by the bolt through the through holes **134** and the first screw holes **124**. The second bump **121b** is provided with a first connecting opening **1203**, a second connecting opening **1204**, and a third connecting opening **1205** along the second direction. The second portion **120b** is provided with a first inserting block **1206**, a second inserting block **1207**, and a third inserting block **1208** respectively adapted to the first connecting opening **1203**, the second connecting opening **1204**, and the third connecting opening **1205**. The first inserting block **1206** cooperates with the first connecting opening **1203**, the second inserting block **1207** cooperates with the second connecting opening **1204**, and the third inserting block **1208** cooperates with the third connecting opening **1205**, to detachably connect the second bump **121b** and the second portion **120b** together. When the wheelbase of the driving device needs to be changed, only the second portion **120b** and the third portion **120c** need to be replaced, reducing the costs.

(26) Referring to FIG. **3**, the driving assembly **100** also includes a support frame **126**, which is connected to the bearing platform **20** in the second direction. The support frame **126** includes a fixing plate **126a** and a support shaft **126b**. The fixing plate **126a** is connected to the bearing platform **20**, one end of the support shaft **126b** is connected with the fixing plate **126a** along the second direction, and the middle part of the cantilever **120** is rotatably supported on the support shaft **126b**. Preferably, the fixing plate **126a** is accommodated in the space enclosed by the first bump **121a**, the second bump **121b**, and the first portion **120a**. The first portion **120a** is rotatably supported on the support shaft **126b**. Further, referring to FIGS. **3** and **4**, the support frame **126** also includes a bearing **125**, which is sleeved outside the support shaft **126b**. The first portion **120a** is provided with a receiving cavity **122**, which passes through the first portion **120a** in the second direction and is located between the first bump **121a** and the second bump **121b**. The bearing **125** is fitted with the receiving cavity **122** to facilitate disassembly and replacement, so that the cantilever **120** and the driving device **13** can rotate around the support shaft **126b**, so that the cantilever assembly **12** provides a suspension system to ensure stability and smooth motion during movement.

(27) In other embodiments, the driving device **13** is fixedly connected with the bearing platform **20** without the support frame **126**, and the cantilever **120** is detachably connected to the driving device **13** through the mounting plate **13b** connected to the driving device **13**.

(28) In this embodiment, referring to FIGS. **3** and **4**, the inner surface of the first portion **120a** is provided with a first annular boss **122c**, and the receiving cavity **122** includes a first cavity **122a** and a second cavity **122b**. The top surface of the support shaft **126b** is provided with a screw **128**. Two bearings **125** are respectively installed in the first cavity **122a** and the second cavity **122b** and are held against the first annular boss **122c**. One of the two bearings **125** is sealed in the second cavity **122b** through a nut **129a** and the screw **128**. Further, a washer **129b** can be added between the nut **129a** and the bearing **125**.

(29) Referring to FIGS. **1** and **7**, this embodiment also provides a transfer device **200**, which includes bearing platform **20** and two driving assemblies **100**. The two driving assemblies **100** are arranged on opposite sides of the bearing platform **20** along the first direction. Wherein, the driving device **13** or the cantilever **120** is connected with the bearing platform **20**. The transfer device **200** can turn through different power outputs of the two driving devices **13** of the two assemblies **100**.

(30) Referring to FIG. **8**, the transfer device **200** also includes a suspension structure. The suspension structure comprises a connecting bridge **21** and a secondary driven wheel **10c** which are rotatably connected to both ends of the connecting bridge **21**. The middle part of the connecting

bridge **21** is rotatably connected with the bearing platform **20**, and the connecting bridge **21** is arranged at the end of the bearing platform **20** opposite to the driven wheel **10b** in the first direction, that is, the driving wheel **10a** is arranged between the driven wheel **10b** and the secondary driven wheel **10c** in the first direction.

(31) Further, the suspension structure also includes a bearing assembly **22**. The bearing assembly **22** is arranged on the connecting bridge **21**, and the bearing assembly **22** is fixedly connected with the bearing platform **20** to rotatably connect the connecting bridge **21** around the bearing assembly **22**. In this embodiment, the bearing assembly **22** may be a combination of the bearing **125** and the support frame **126**.

(32) In this embodiment, the driving component **13a** may be an electric motor and the driving wheel **10a** may be a plastic or rubber tire. The driven wheel **10b** and the secondary driven wheel **10c** may be omni-directional wheels **10d** that may be driven to rotate when pointing anywhere through 360 degrees direction of motion of platform.

(33) The transfer device **200** also includes a control module (not shown in the drawings) and a sensor module **23**. The control module is used to control the startup and output power of the driving device **13**. The sensor module **23** is used to sense the speed of the driving wheel **10a**, and the sensor module **23** is installed on the driving wheel **10a**.

(34) The above embodiments are only used to illustrate the technical solution of the application rather than for limitation. Although the application is described in detail with reference to the above preferred embodiments, those skilled in the art should understand that any modification or equivalent replacement of the technical solution of the application should not deviate from the spirit and scope of the technical solution of the application.

Claims

1. A driving assembly for driving a bearing platform, comprising: a driving device; a driving wheel connected to the driving device, rotatable by the driving device; a cantilever assembly comprising a cantilever configured for mounting to the bearing platform, and, a driven wheel rotatably connected to one end of the cantilever, and the other end of the cantilever is detachably connected to a fixed part of the driving device; wherein the driving device further comprises a driving component and a mounting plate; the mounting plate comprises opposite first and second ends, the first end is fixedly connected with the driving component, and the second end protrudes from the driving component along a first direction; wherein, the cantilever comprises a first portion, a second portion and a third portion in the first direction; wherein the first portion is connected with the second end, the third portion is connected with the driven wheel, the second portion is connected between the first portion and the third portion, the third portion and the first portion extend along the first direction respectively and are arranged at parallel intervals in the third direction, wherein the third direction is perpendicular to the first direction; wherein a first bump is provided at one end of the first portion along the second direction, a second bump is provided at another end of the first portion along the second direction, the first bump and the second bump are arranged on the same side of the first portion along the first direction, and the second direction is perpendicular to the first direction; the first bump is detachably connected with the second end along the second direction.

2. The driving assembly according to claim 1, wherein a plurality of positioning columns and a plurality of first screw holes are provided at top surface along the second direction of the first bump; the mounting plate is provided with a plurality of positioning holes matched with the positioning column and a plurality of through holes corresponding to the first screw hole; the first bump is detachably connected to the second portion by bolts passing through the first screw holes and the through holes.

3. The driving assembly according to claim 1, wherein the second bump is provided with a first

connecting opening, a second connecting opening and a third connecting opening along the second direction; the second portion is provided with a first inserting block, a second inserting block and a third inserting block respectively matched to the first connecting opening, the second connecting opening and the third connecting opening; the first inserting block is inserted into the first connecting opening, the second inserting block is inserted into the second connecting opening, and the third inserting block is inserted into the third connecting opening, to connect the second bump and the second portion together.

4. The driving assembly according to claim 1, wherein the driving assembly further comprises a bearing frame connected to the bearing platform along the second direction; the bearing frame comprises a fixing plate and a support shaft, the fixing plate is connected to the bearing platform, one end of the support shaft is connected with the fixing plate along the second direction, and middle part of the cantilever is rotatably supported on the support shaft.

5. A transfer device, wherein the transfer device comprises the bearing platform and two driving assemblies according to claim 1, the two driving assemblies are arranged on opposite sides of the bearing platform along the first direction, wherein the two driving assemblies or the cantilever is connected with the bearing platform.

6. The transfer device according to claim 5, wherein a plurality of positioning columns and a plurality of first screw holes are provided at top surface along the second direction of the first bump; the mounting plate is provided with a plurality of positioning holes matched with the positioning column and a plurality of through holes corresponding to the first screw hole; the first bump is detachably connected to the second portion by bolts passing through the first screw holes and the through holes.

7. The transfer device according to claim 5, wherein the second bump is provided with a first connecting opening, a second connecting opening and a third connecting opening along the second direction; the second portion is provided with a first inserting block, a second inserting block and a third inserting block respectively matched to the first connecting opening, the second connecting opening and the third connecting opening; the first inserting block is inserted into the first connecting opening, the second inserting block is inserted into the second connecting opening, and the third inserting block is inserted into the third connecting opening, to connect the second bump and the second portion together.

8. The transfer device according to claim 5, wherein the driving assembly further comprises a bearing frame connected to the bearing platform along the second direction; the bearing frame comprises a fixing plate and a support shaft, the fixing plate is connected to the bearing platform, one end of the support shaft is connected with the fixing plate along the second direction, and middle part of the cantilever is rotatably supported on the support shaft.

9. The transfer device according to claim 5, wherein the transfer device further comprises a connecting bridge and two secondary driven wheels, the two secondary driven wheels are rotatably connected to both ends of the connecting bridge respectively, a middle of the connecting bridge is rotatably connected to the bearing platform, and the two secondary driven wheels are located at an end of the bearing platform opposite to the driven wheel along the first direction.

10. The transfer device according to claim 9, wherein the driven wheel and the secondary driven wheel are omni-directional wheels.
